

Dynamic Column Names in R: Seven Approaches Compared

Ronald G. Thomas

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Introduction

While reviewing some production R code recently, the following pattern appeared:

```
data |>
  mutate(
    !!paste0(measure, "_bl") := baseline_value,
    !!paste0(measure, "_cng") := current - baseline_value
  )
```

The `!!` and `:=` operators were unfamiliar. After investigation, they proved to be part of R's tidy evaluation system: a powerful but often misunderstood feature for programmatic data manipulation.

This exploration led to cataloguing seven different approaches to creating columns with dynamic names in R:

1. Base R using `[]` assignment
2. Base R using `do.call()` and `setNames()`
3. The “classic” tidyverse approach with `!!` and `:=`
4. The modern tidyverse glue-style syntax

5. The rlang expression splicing pattern
6. The data.table approach
7. The collapse package approach

The Problem

Consider a function that calculates change scores for any cognitive measure. Given a dataframe with columns `rid` (subject ID), `vis` (visit), and a measure like `mmse`, we wish to create two new columns:

- `mmse_bl`: the baseline value for each subject
- `mmse_cng`: the change from baseline at each visit

The column names must be constructed dynamically because the function should work for any measure (`mmse`, `adas13`, `cdr`, etc.).

Here is our sample data:

```
library(dplyr)

df <- tibble(
  rid = rep(c("001", "002"), each = 3),
  vis = rep(c("bl", "m06", "m12"), 2),
  mmse = c(28, 27, 25, 30, 29, 28)
)

df
```

```
# A tibble: 6 x 3
  rid   vis   mmse
<chr> <chr> <dbl>
1 001   bl     28
2 001  m06     27
3 001  m12     25
4 002   bl     30
5 002  m06     29
6 002  m12     28
```

Approach 1: Base R with `[]` Assignment

The most straightforward approach uses base R's `[]` assignment operator:

```

calculate_change_base <- function(data, measure) {
  bl_col <- paste0(measure, "_bl")
  cng_col <- paste0(measure, "_cng")

  result <- data

  for (id in unique(data$rid)) {
    idx <- data$rid == id
    baseline <- data[[measure]][idx & data$vis == "bl"][1]
    result[[bl_col]][idx] <- baseline
    result[[cng_col]][idx] <- data[[measure]][idx] - baseline
  }

  result
}

calculate_change_base(df, "mmse")

```

```

# A tibble: 6 x 5
  rid   vis   mmse mmse_bl mmse_cng
  <chr> <chr> <dbl>   <dbl>   <dbl>
1 001   bl     28     28     0
2 001  m06     27     28    -1
3 001  m12     25     28    -3
4 002   bl     30     30     0
5 002  m06     29     30    -1
6 002  m12     28     30    -2

```

Discussion

How it works: The `[[` operator accepts character strings for column names, unlike `$` which requires literal names. Writing `df[["mmse"]]` is equivalent to `df$mmse`, but the former allows `df[[variable]]` where `variable` contains the column name.

Advantages:

- No dependencies beyond base R
- Syntax is familiar to programmers from other languages
- Explicit and easy to debug

Disadvantages:

- Verbose, especially with grouped operations
- Requires explicit loops for by-group calculations
- Mutable state pattern (`result` is modified in place)
- Does not integrate with dplyr pipelines
- Performance degrades with many groups

When to use: Small scripts with no package dependencies, or when interfacing with code from other languages where this pattern is standard.

Approach 2: Base R with `do.call()` and `setNames()`

A more functional base R approach avoids explicit loops:

```
calculate_change_docal1 <- function(data, measure) {
  bl_col <- paste0(measure, "_bl")
  cng_col <- paste0(measure, "_cng")

  baselines <- tapply(
    data[[measure]][data$vis == "bl"],
    data$rid[data$vis == "bl"],
    FUN = `[`, 1
  )

  bl_values <- baselines[data$rid]
  cng_values <- data[[measure]] - bl_values

  new_cols <- setNames(
    list(as.numeric(bl_values), as.numeric(cng_values)),
    c(bl_col, cng_col)
  )

  do.call(cbind, c(list(data), new_cols))
}

calculate_change_docal1(df, "mmse")
```

	rid	vis	mmse	mmse_bl	mmse_cng
1	001	bl	28	28	0
2	001	m06	27	28	-1
3	001	m12	25	28	-3
4	002	bl	30	30	0
5	002	m06	29	30	-1
6	002	m12	28	30	-2

Discussion

How it works: `setNames()` creates a named list where the names come from a character vector. `do.call(cbind, ...)` then binds these columns to the original dataframe. The `tapply()` function computes grouped summaries without explicit loops.

Advantages:

- No external dependencies
- Functional style (no mutable state)
- Vectorized operations for better performance

Disadvantages:

- Dense, less readable syntax
- `tapply()` returns an array requiring careful indexing
- Type coercion issues (note the `as.numeric()` calls)
- Returns a `data.frame`, not a `tibble`

When to use: Package development where minimising dependencies is critical, or performance-sensitive code where the overhead of `dplyr` is unacceptable.



Figure 1: Tidy evaluation approaches.

Approach 3: Classic Tidyverse with `!!` and `:=`

The tidyverse introduced tidy evaluation to handle dynamic column names. Two operators are central:

- `:=` (the “walrus” operator): Allows the left-hand side of an assignment to be evaluated
- `!!` (bang-bang): Unquotes an expression, forcing immediate evaluation

```

calculate_change_classic <- function(data, measure) {
  bl_col <- paste0(measure, "_bl")
  cng_col <- paste0(measure, "_cng")

  data |>
    group_by(rid) |>
    mutate(
      !!bl_col := .data[[measure]][vis == "bl"][1],
      !!cng_col := .data[[measure]] - .data[[bl_col]]
    ) |>
    ungroup()
}

calculate_change_classic(df, "mmse")

```

```

# A tibble: 6 x 5
  rid   vis   mmse mmse_bl mmse_cng
  <chr> <chr> <dbl>   <dbl>   <dbl>
1 001   bl     28     28     0
2 001  m06     27     28    -1
3 001  m12     25     28    -3
4 002   bl     30     30     0
5 002  m06     29     30    -1
6 002  m12     28     30    -2

```

Discussion

How it works: Standard `mutate()` syntax like `mutate(new_col = value)` treats `new_col` as a literal name. The `=` operator does not evaluate its left-hand side. The `:=` operator changes this behaviour. When `mutate(!!bl_col := value)` is written, the `!!` forces `bl_col` to be evaluated (yielding `"mmse_bl"`), and `:=` uses that string as the column name.

The `.data` pronoun explicitly references columns in the dataframe, avoiding ambiguity between dataframe columns and local variables.

Advantages:

- Integrates seamlessly with dplyr pipelines
- Grouped operations are trivial
- Declarative, readable intent (once the syntax is learnt)

Disadvantages:

- Unfamiliar syntax for newcomers (`!!` and `:=` are not standard R)
- Requires understanding tidy evaluation concepts
- The `rlang` package must be available (loaded with `dplyr`)

When to use: Legacy tidyverse code (pre-dplyr 1.0), or when the full power of quasiquotation is needed for complex metaprogramming.

Approach 4: Modern Tidyverse with Glue Syntax

Starting with dplyr 1.0 (June 2020), a cleaner syntax emerged using glue-style interpolation:

```
calculate_change_modern <- function(data, measure) {
  data |>
    group_by(rid) |>
    mutate(
      "{measure}_bl" := .data[[measure]][vis == "bl"][1],
      "{measure}_cng" := .data[[measure]] - .data[[paste0(measure, "_bl")]]
    ) |>
    ungroup()
}

calculate_change_modern(df, "mmse")
```

```
# A tibble: 6 x 5
  rid   vis   mmse mmse_bl mmse_cng
<chr> <chr> <dbl>   <dbl>   <dbl>
1 001   bl     28     28     0
2 001 m06     27     28    -1
3 001 m12     25     28    -3
4 002   bl     30     30     0
5 002 m06     29     30    -1
6 002 m12     28     30    -2
```

Discussion

How it works: The string "{measure}_bl" is interpolated like `glue::glue()`, substituting the value of `measure` directly. This occurs at the level of the column name, with `:=` still required to enable left-hand side evaluation.

Advantages:

- Most readable of all tidyverse approaches
- Reduces cognitive load (no need to remember !! semantics)
- Consistent with glue syntax used elsewhere in the tidyverse

Disadvantages:

- Requires dplyr $\geq$ 1.0
- Still requires `:=` (cannot use `=`)

- Less flexible than `!!` for complex quasiquotation

When to use: New tidyverse code. This is the current recommended approach for dynamic column names in dplyr.

Approach 5: rlang Expression Splicing

For more explicit metaprogramming, rlang provides `expr()` for building expressions and `!!!` for splicing them into function calls:

```
library(rlang)

calculate_change_rlang <- function(data, measure) {
  bl_col <- paste0(measure, "_bl")
  cng_col <- paste0(measure, "_cng")

  exprs <- list(
    expr(.data[!!measure][vis == "bl"][1]),
    expr(.data[!!measure] - .data[!!bl_col])
  )
  names(exprs) <- c(bl_col, cng_col)

  data |>
    group_by(rid) |>
    mutate(!!!exprs) |>
    ungroup()
}

calculate_change_rlang(df, "mmse")
```

```
# A tibble: 6 x 5
  rid   vis   mmse mmse_bl mmse_cng
<chr> <chr> <dbl>   <dbl>   <dbl>
1 001   bl     28     28     0
2 001  m06     27     28    -1
3 001  m12     25     28    -3
4 002   bl     30     30     0
5 002  m06     29     30    -1
6 002  m12     28     30    -2
```


Discussion

How it works: `expr()` captures an expression without evaluating it, while `!!` inside `expr()` forces evaluation of specific parts. The result is a list of named expressions. The `!!!` (splice) operator unpacks this list into `mutate()`, equivalent to writing each expression as a separate argument.

Advantages:

- Build expressions programmatically before evaluation
- Useful when the number of columns is dynamic
- Expressions can be inspected, modified, or logged before use
- Clear separation between expression construction and evaluation

Disadvantages:

- More verbose for simple cases
- Requires understanding quasiquotation deeply
- Overkill for straightforward dynamic naming

When to use: Complex metaprogramming where expressions must be built dynamically, such as generating an unknown number of columns based on input data, or when inspecting or logging expressions before evaluation is desirable.



Figure 2: High-performance alternatives.

Approach 6: `data.table`

The `data.table` package has its own `:=` operator that predates tidyverse adoption:

```
library(data.table)

calculate_change_dt <- function(data, measure) {
  bl_col <- paste0(measure, "_bl")
  cng_col <- paste0(measure, "_cng")

  dt <- as.data.table(data)

  dt[, (bl_col) := .SD[[measure]][vis == "bl"][1], by = rid]
  dt[, (cng_col) := .SD[[measure]] - .SD[[bl_col]], by = rid]

  dt[]
}

calculate_change_dt(df, "mmse")
```

	rid	vis	mmse	mmse_bl	mmse_cng
	<char>	<char>	<num>	<num>	<num>
1:	001	bl	28	28	0
2:	001	m06	27	28	-1
3:	001	m12	25	28	-3
4:	002	bl	30	30	0
5:	002	m06	29	30	-1
6:	002	m12	28	30	-2

Discussion

How it works: In `data.table`, wrapping a variable in parentheses on the left-hand side of `:=` forces evaluation: `(bl_col)` evaluates to `"mmse_bl"`. Without parentheses, `bl_col := value` would create a column literally named `"bl_col"`. The `.SD` pronoun (Subset of Data) references the current group's data.

Advantages:

- Extremely fast, especially for large datasets
- Memory efficient (modifies in place by reference)
- Mature, battle-tested codebase
- The `by` argument handles grouping concisely

Disadvantages:

- Different mental model from base R and tidyverse
- Modify-by-reference semantics can cause subtle bugs
- Less readable for those unfamiliar with `data.table` idioms
- Parentheses syntax `(col)` is easy to forget

When to use: Performance-critical applications, very large datasets (millions of rows), or codebases already using `data.table`.



Figure 3: Minimalist high-speed data tools.

Approach 7: collapse

The collapse package offers high-performance data manipulation with its own idioms:

```
library(collapse)

calculate_change_collapse <- function(data, measure) {
  bl_col <- paste0(measure, "_bl")
  cng_col <- paste0(measure, "_cng")

  bl_data <- fsubset(data, vis == "bl")
  bl_values <- get_vars(bl_data, c("rid", measure))
  names(bl_values)[2] <- bl_col

  result <- join(data, bl_values, on = "rid", how = "left")
  result[[cng_col]] <- result[[measure]] - result[[bl_col]]

  result
}
```

```
}
calculate_change_collapse(df, "mmse")
```

```
left join: data[rid] 6/6 (100%) <3:1st> bl_values[rid] 2/2 (100%)
```

```
# A tibble: 6 x 5
  rid   vis   mmse mmse_bl mmse_cng
  <chr> <chr> <dbl>   <dbl>   <dbl>
1 001   bl     28     28     0
2 001  m06     27     28    -1
3 001  m12     25     28    -3
4 002   bl     30     30     0
5 002  m06     29     30    -1
6 002  m12     28     30    -2
```

Discussion

How it works: collapse provides fast versions of common operations (`fsubset`, `get_vars`, `join`, etc.). Unlike `dplyr`, collapse does not use tidy evaluation, so dynamic column access requires base R syntax like `get_vars()` with character vectors and direct `[]` assignment. The approach above uses a join strategy rather than grouped mutation.

Advantages:

- Extremely fast (often faster than `data.table` for certain operations)
- Low memory footprint
- Works with both `data.frames` and `data.tables`
- Good for time series and panel data

Disadvantages:

- Smaller user community than tidyverse or `data.table`
- Inconsistent tidy evaluation support across functions
- Often requires mixing collapse functions with base R syntax
- Less comprehensive documentation

When to use: Performance-critical code, especially time series or panel data analysis. Also useful when speed is needed but `data.table`'s reference semantics are to be avoided.

Comparison Summary

Approach	Dependencies	Readability	Performance	Best For
Base R []	None	Moderate	Low	No-dependency scripts
Base R do.call	None	Low	Moderate	Package development
Classic !! :=	dplyr	Low	Moderate	Legacy tidyverse code
Modern glue	dplyr >= 1.0	High	Moderate	New tidyverse code
rlang !!! splice	rlang	Moderate	Moderate	Complex metaprogramming
data.table	data.table	Moderate	High	Large datasets
collapse	collapse	Moderate	Very High	Performance-critical work

Practical Example: Multiple Measures

The real power emerges when processing multiple measures:

```
df_multi <- tibble(
  rid = rep(c("001", "002"), each = 3),
  vis = rep(c("bl", "m06", "m12"), 2),
  mmse = c(28, 27, 25, 30, 29, 28),
  adas = c(12, 14, 18, 10, 12, 15),
  cdr = c(0.5, 0.5, 1.0, 0.5, 0.5, 0.5)
)

calculate_all_changes <- function(data, measures) {
  result <- data |> group_by(rid)

  for (measure in measures) {
    result <- result |>
      mutate(
        "{measure}_bl" := .data[[measure]][vis == "bl"][1],
        "{measure}_cng" := .data[[measure]] - .data[[paste0(measure, "_bl")]]
      )
  }

  result |> ungroup()
}

calculate_all_changes(df_multi, c("mmse", "adas", "cdr"))
```

```
# A tibble: 6 x 11
```

	rid	vis	mmse	adas	cdr	mmse_bl	mmse_cng	adas_bl	adas_cng	cdr_bl	cdr_cng
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	001	bl	28	12	0.5	28	0	12	0	0.5	0
2	001	m06	27	14	0.5	28	-1	12	2	0.5	0
3	001	m12	25	18	1	28	-3	12	6	0.5	0.5
4	002	bl	30	10	0.5	30	0	10	0	0.5	0
5	002	m06	29	12	0.5	30	-1	10	2	0.5	0
6	002	m12	28	15	0.5	30	-2	10	5	0.5	0

Common Pitfalls

1. Forgetting := on the LHS (tidyverse)

```
# Wrong: creates column literally named "{measure}_bl"
mutate("{measure}_bl" = value)

# Correct: evaluates the string
mutate("{measure}_bl" := value)
```

2. Forgetting parentheses (data.table)

```
# Wrong: creates column named "bl_col"
dt[, bl_col := value]

# Correct: evaluates bl_col to get "mmse_bl"
dt[, (bl_col) := value]
```

3. Confusing column references

```
# Ambiguous: is 'measure' a column or variable?
mutate("{measure}_bl" := measure[vis == "bl"][1])

# Clear: .data[[measure]] references the column
mutate("{measure}_bl" := .data[[measure]][vis == "bl"][1])
```

4. Reference semantics in data.table

```
dt <- as.data.table(df)
dt2 <- dt # This is NOT a copy!
dt2[, new_col := 1] # Modifies both dt and dt2

# Safe copy:
dt2 <- copy(dt)
```

What Did We Learn?

Lessons Learnt

1. **:= enables dynamic LHS:** In both tidyverse and data.table, this operator allows the left side of an assignment to be evaluated
2. **!! is the unquote operator:** Forces immediate evaluation of an expression in tidy eval contexts
3. **Glue syntax is cleaner:** "{var}_suffix" is more readable than !!paste0(var, "_suffix")
4. **Parentheses matter in data.table:** (col) evaluates, col is literal
5. **.data removes ambiguity:** Always use .data[[col]] for column references in functions
6. **Choose based on context:** Readability (glue) vs performance (data.table) vs no dependencies (base R)

Related posts in this cluster

This post is part of the *R Language and Metaprogramming* series. Recommended reading order:

1. Post 62: [The Pipe Equivalence Myth](#)
2. **Post 63: Dynamic Column Names: Seven Approaches Compared** (this post)

Reproducibility

```
sessionInfo()
```

R version 4.6.1 (2026-06-24)

Platform: aarch64-apple-darwin25.4.0

Running under: macOS Tahoe 26.6.2

Matrix products: default

BLAS: /opt/homebrew/Cellar/openblas/0.3.34/lib/libopenblas-r0.3.34.dylib

LAPACK: /opt/homebrew/Cellar/r/4.6.1/lib/R/lib/libRlapack.dylib; LAPACK version 3.12.1

locale:

[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

```
time zone: America/Los_Angeles
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods   base
```

```
other attached packages:
```

```
[1] collapse_2.1.7    data.table_1.18.4  rlang_1.3.0       dplyr_1.2.1
```

```
loaded via a namespace (and not attached):
```

```
[1] vctrs_0.7.3      cli_3.6.6         knitr_1.51        xfun_0.58
[5] otel_0.2.0       generics_0.1.4    jsonlite_2.0.0    glue_1.8.1
[9] htmltools_0.5.9  rmarkdown_2.31    evaluate_1.0.5    tibble_3.3.1
[13] fastmap_1.2.0    yaml_2.3.12       lifecycle_1.0.5   compiler_4.6.1
[17] Rcpp_1.1.1-1.1   pkgconfig_2.0.3   digest_0.6.39     R6_2.6.1
[21] tidyselect_1.2.1 utf8_1.2.6        pillar_1.11.1     parallel_4.6.1
[25] magrittr_2.0.5   tools_4.6.1
```

Let's Connect

Have questions, suggestions, or spot an error? Let me know.

- **GitHub:** [rgt47](#)
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I would enjoy hearing from readers who:

- Spot an error or a better approach to any of the code in this post.
- Have suggestions for topics to cover.
- Want to discuss R programming, data science, or reproducible research.
- Have questions about anything in this tutorial.
- Simply want to say hello and connect.